

Name: _____

Period: _____

Date: _____

Topic 2.5 Guided Notes: Storing Memories

Unit 2: Cognition | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appspsychreviewhub.com/overviews/topic2.5

Word Bank

<i>Retrograde Amnesia</i>	<i>Alzheimer's Disease</i>	<i>Short-Term Memory</i>
<i>Working Memory</i>	<i>Sensory Memory</i>	<i>Elaborative Rehearsal</i>
<i>Anterograde Amnesia</i>	<i>Autobiographical Memory</i>	<i>Infantile Amnesia</i>
<i>Maintenance Rehearsal</i>	<i>Long-Term Memory</i>	<i>Highly Superior Autobiographical Memory (HSAM)</i>

1. The Four Storage Systems: Duration, Capacity, and Content

Before rehearsal or amnesia can make sense, it helps to see how the four memory stores actually differ from each other. Each one holds a different amount of information, for a different length of time, holding a different kind of content:

- _____: holds an almost complete, high-capacity snapshot of raw sensory input (sights, sounds), but only for a fraction of a second to a few seconds before it fades.
- _____: holds a small handful of items — classically around seven — for roughly 15 to 30 seconds unless you actively rehearse them.
- _____: covers similar territory to short-term memory in duration and capacity, but describes the active manipulation of that information (solving a problem, holding a conversation) rather than just passively holding onto it.
- _____: once information survives long enough to be encoded, it can be held here with a virtually unlimited capacity for anywhere from minutes to an entire lifetime.

Apply It: Which Storage System?

Using the word bank above, identify which concept best matches each scenario.

#	Scenario	Concept
1	A witness to a car accident can only hold the license plate number in mind for about 20 seconds unless she keeps repeating it to herself.	
2	While solving a math problem in her head, a student mentally juggles several numbers at once, actively working with them rather than just holding them.	
3	Decades after graduating, a woman can still recall her college roommate's name without any effort.	

2. Keeping Memories Alive: The Two Types of Rehearsal

If you want a memory to survive beyond the brief window of short-term or working memory, you have to practice it. However, the type of practice you choose determines how long the memory will last.

- : this involves simply repeating information over and over in your head to keep it active in your short-term memory. If you get a 6-digit verification code texted to your phone, you might mutter the digits repeatedly until you type it in. The second you type it, the memory vanishes. It doesn't encode meaning, so it rarely makes it to long-term storage.
- : this involves actively thinking about the meaning of the new information and connecting it to existing knowledge you already have stored. Instead of just repeating an AP Psych vocabulary word, you think of a personal example of when you experienced it. This builds a web of strong neural connections, safely securing the information in your long-term memory.

Apply It: Which Type of Rehearsal?

#	Scenario	Concept
1	A man repeats a new coworker's name silently to himself the moment they are introduced, but forgets it entirely five minutes later.	
2	A student learning about the Sunk-Cost Fallacy immediately thinks of the time she refused to leave a boring movie because she'd already paid for the ticket.	

3. The Story of You

Our brains are inherently selfish; we remember things much better when they relate directly to us. Your is the collection of personal life events and knowledge about yourself that forms your identity. It is a powerful blend of episodic memories (like your first kiss) and semantic memories (like knowing you were born in 2008).

While most of us have a standard autobiographical memory that fades over time, there is a fascinating, rare biological condition known as . People with this condition can recall an extremely detailed, accurate account of nearly every single day of their lives, often down to the weather, what they ate for lunch, and the exact day of the week a date fell on decades ago, all without any conscious effort or mnemonic devices!

Apply It: Autobiographical Memory or HSAM?

#	Scenario	Concept
1	When asked what she did exactly nine years ago on a random Tuesday, a woman instantly and effortlessly recalls the specific weather, her outfit, and a conversation she had that day.	
2	A man's sense of identity is built from a blend of specific personal memories, like his wedding day, and general facts about himself, like his own birthdate.	

4. When the Filing Cabinet Breaks: Amnesia and Disease

Memory storage is a fragile biological process. Physical trauma, disease, and developmental stages can severely impair our ability to hold onto information.

- _____: why can't you remember your first birthday party? It isn't because the memories faded; it's because your brain literally didn't have the hardware to store them yet! The hippocampus (the brain structure crucial for forming explicit memories) isn't fully developed until around age 3 or 4.
- _____: a type of memory loss where a person cannot recall events that occurred before the injury or trauma. They lose their past (like forgetting their name, their family, or their childhood), but they can still form new memories moving forward.
- _____: a type of memory loss where a person cannot form new long-term explicit memories after the injury, though their past memories remain intact. This was famously documented in the case study of Patient H.M., whose hippocampus was surgically removed. Every time he met his doctor, it was as if they were meeting for the very first time, even if they had spoken just five minutes earlier.
- _____: a progressive, fatal neurological disorder that destroys brain cells, causing severe memory loss and cognitive decline. It typically attacks the areas of the brain involved in memory first, meaning recent memories are lost before older, deeply consolidated memories.

Apply It: Which Type of Memory Loss?

#	Scenario	Concept
1	After a car accident, a woman can no longer recall anything about the months leading up to the crash, but she has no trouble learning and remembering new information since waking up.	
2	Following brain surgery, a man can hold a completely normal conversation, but 10 minutes later has no memory the conversation ever happened, though he still clearly remembers his life before the operation.	
3	No adult can consciously recall being breastfed as an infant, no matter how hard they try.	
4	An elderly man in the later stages of a progressive brain disease can vividly describe his childhood home in detail but cannot remember that his wife visited him an hour ago.	

5. Don't Trip Up! (Common Misconceptions)

Remember the prefixes: "Retro-" means backward or past (like "retro" clothing). You lose the memories prior to the injury — this is _____. "Antero-" means forward. You can't move forward to create new memories — this is _____.

_____ just maintains the memory in your short-term storage. _____ encodes it deeply into long-term storage!

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. Rather than simply repeating a new AP Psychology vocabulary word over and over, a student thinks about what the term actually means and connects it to a specific personal experience from her own life. This deeper strategy, which builds strong neural connections for long-term storage, is called:
 - A. Sensory Memory
 - B. Massed Practice
 - C. Elaborative Rehearsal
 - D. Maintenance Rehearsal
2. A college student has absolutely no memory of his very first birthday party, not because the memory faded away, but because his hippocampus simply wasn't developed enough at that young age to properly store the explicit memory in the first place. This inability to recall very early childhood events is called:
 - A. Anterograde Amnesia
 - B. Infantile Amnesia
 - C. HSAM
 - D. Retrograde Amnesia
3. After a severe head injury, a man can no longer recall his own childhood, his wedding day, or even his own name, yet doctors confirm he is still fully capable of learning and remembering new information from this point forward. This specific inability to recall memories from before the injury is called:
 - A. Anterograde Amnesia
 - B. HSAM
 - C. Infantile Amnesia
 - D. Retrograde Amnesia
4. Famously documented in the case of Patient H.M., whose hippocampus was surgically removed, a specific type of amnesia leaves old memories completely intact, but makes a person unable to form any brand new long-term memories going forward, meaning they might meet the same doctor over and over as if for the very first time. This is:
 - A. Highly Superior Autobiographical Memory
 - B. Infantile Amnesia
 - C. Anterograde Amnesia
 - D. Retrograde Amnesia
5. A woman in the early stages of a progressive, fatal neurological disease struggles to remember what she had for breakfast that same morning, yet she can still vividly recall detailed memories from her childhood many decades ago. Her disease typically attacks recent memories before older, more deeply consolidated ones. This disease is:
 - A. Anterograde Amnesia
 - B. Infantile Amnesia
 - C. Retrograde Amnesia
 - D. Alzheimer's Disease



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>

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